

ING 3 IA--- Mobile Applications Practicum

Practical 1 : Flutter IDE discovering

1) Introduction :

- This first practical is an introduction to the two main integrated development environment (IDE) for Flutter applications, namely *Android Studio & Visual Code Studio (VScode)*. Its goal is to become familiar with these IDEs and learn how to integrate and use Dart & Flutter packages.

2) Creating an app :

- Launch your IDE then create a new Flutter project using the context menu, the terminal or the command palette (Vs Code).
- Explore and examine the different parts of the interface (the menu bar, the execution toolbar, the status bar, the panel, the project explorer, etc.). Ref. Figure 1 & Figure 2.
- Discover the structure of a Flutter project.

3) Running the app :

- Launch an Android emulator (iOS simulator) or connect your physical device.
- Run the app using the context menu, the toolbar, the keyboard shortcut or the terminal.
- Discover the debug mode as well as the *Hot Reload* option.

4) Dart/Flutter packages integration :

- Discover the default app i.e. "*Counter App*".
- Using a package (e.g. *url_launcher*) :

- a) Download the **url_launcher** by adding the dependency **"url_launcher: ^6.2.4"** in the **"pubspec.yaml"** file under the **"dependencies"** part. Be careful to respect the indentation.

```
dependencies:  
  flutter:  
    sdk: flutter  
  url_launcher: ^6.2.4
```

NB : It's also possible to download the package by executing the command **"\$ flutter pub add url_launcher"** from the terminal.

- b) Install the **url_launcher** package in your "Counter" application by running the **"flutter pub get"** command.
- c) Import the package from the **"main.dart"** file by including the line : **"import 'package:url_launcher/url_launcher.dart';"**
- d) In the **"main.dart"** file, replace the **"_incrementCounter()"** function with the following function :

```
void launchUrlFct( ) async {  
    await launchUrl (Uri ( scheme : 'https', host : 'www.google.com'),  
                      mode : LaunchMode.externalApplication) ;  
}
```

- e) Associate the **"launchUrlFct"** function to the floating button. To do this, locate the floating button code (floatingActionButton) then, in the onPressed method, replace the **"_incrementCounter"** function call with **"launchUrlFct"** function call.
- f) Re-run and test the application.

5) Additional part : open different types of links from the same floating button.

- Phone call: *scheme* = 'tel' et *path* = "+213-560-741-723".
- E-mail : *scheme* = 'mailto' et *path* = "contact@gmail.com".
- SMS : *scheme* = 'sms' et *path* = "+213-560-741-723".
- File : `Uri.file("D:\\MyFolder").` // Only for Desktop app.

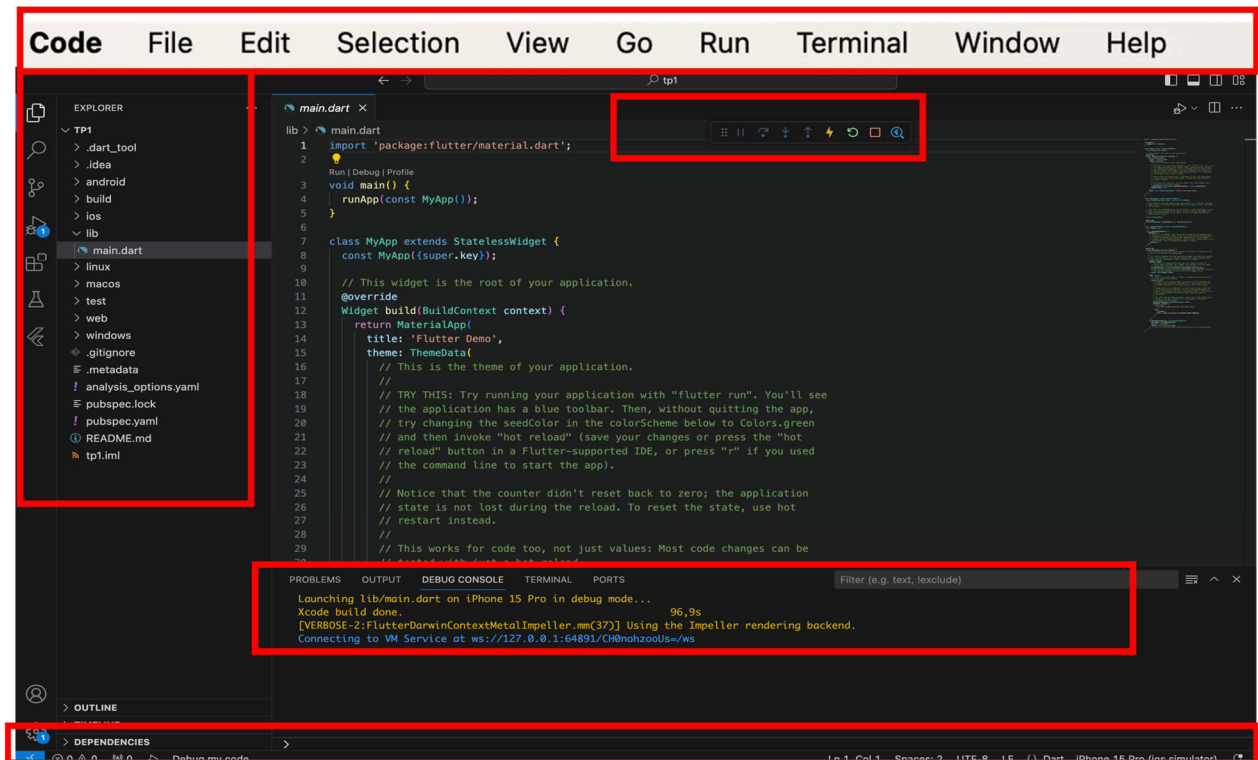


Figure 1: User interface of VS Code (VS Code UI)

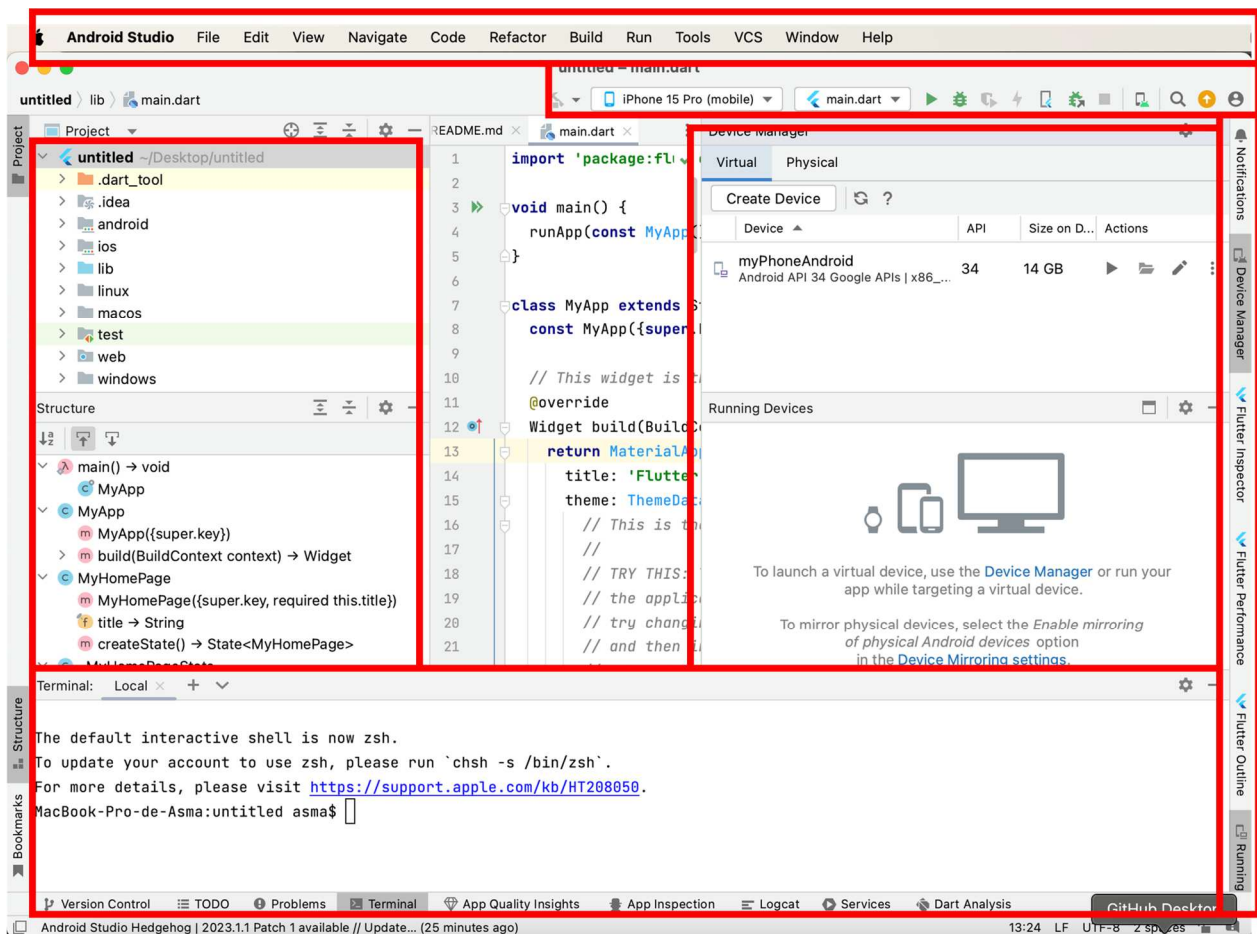


Figure 2: User interface of Android Studio (Android Studio UI)